



An unsupervised subdomain adaptation of cross-domain remaining useful life prediction for sensor-equipped equipments

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ABSTRACT

Most existing remaining useful life (RUL) prediction models assume a similarity between the data distributions in the source and target domains, but this assumption is often challenging in practical applications. Although traditional transfer learning methods could relax this assumption by minimizing the distributional differences between domains, they often ignore the fine-grained features exhibited by equipment at different health state and the inherent properties of the domains. Additionally, most models ignore the importance of the location information of the sensors equipped on the equipment, and the real-world problem of the high cost of fully labeling the data. Addressing these challenges, we introduce a novel model of unsupervised subdomain adaptation and feature disentanglement for equipment RUL prediction. The model contains four components: a feature extractor, a feature similarity component, a domain adaptation component, and a multi-task module. Specifically, the feature extractor designs a dynamic fusion network to divide the domain's private and shared features. The feature similarity component employs a cross-attention mechanism to constrain the extracted features. The domain adaptation enables the proposed model to extract deeper subdomain-shared features. The multi-task module simultaneously divides subdomains and performs RUL prediction. The proposed model is validated using multiple tasks and compared with existing works.

1. Introduction

Sensor-equipped equipment (i.e., sensors installed at critical locations on the equipment to collect data on the operating conditions of the equipment) plays an important role in aerospace, rail transportation, power, and many other industrial applications (Li, Li, Wang, Liu, & Tan, 2024; Zhuang, Jia, Huang, Beer, & Feng, 2024). Failure of this kind of equipments would disrupt an enterprise's production schedule and result in substantial economic losses. Traditional maintenance strategies (e.g., periodic maintenance and breakdown maintenance) are challenging in meeting the demand for high reliability and safety of equipment for industrial applications, and there is a phenomenon of under-maintenance and over-maintenance. Different from traditional maintenance strategies, prognostics and health management (PHM) techniques can make maintenance strategies based on the equipment's health status. Remaining useful life (RUL) prediction, playing an important role in PHM, has recently received much attention from scholars (Cao, Ding, Jia, & Tian, 2021; Lv, Xia, Cheng, Yan, & Wu, 2024; Zhu, Zhou, Li, & Yan, 2024) (RUL denotes the time span between the

equipment's current and failure state). Accurate RUL prediction reduces unnecessary maintenance and minimizes potential risks, thereby improving the safety and reliability of the equipment.

The methods of RUL prediction mainly include physical model-based approaches and data-driven approaches. Compared with physical model-based approaches, data-driven approaches are more widely used in RUL prediction (Deng, Du, Wang, Shao, & Huang, 2023; She & Jia, 2021; Zhuang, Xu, Wang, & Tang, 2024). Deep learning model-based methods, an essential branch of data-driven approaches, have received extensive attention from researchers (Du, Liu, Huang, & Li, 2017; Zhang et al., 2022). Deep learning methods can adaptively construct a non-linear mapping relationship between the captured data and RULs (Li, Wang, et al., 2024). Zhang, Su, Wu, Liu, and Xie (2024) embedded the trend augmentation and temporal feature attention modules into the traditional transformer deep learning model, enhancing its precision in RUL prediction. Considering the gradual degradation of the equipment, Shi et al. (2024) used a lightweight framework integrating long short-term memory (LSTM) with a dual attention strategy to improve

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RUL prediction. Zheng, He, Chen, and Pu (2022) introduced a framework using hybrid dilated convolutional networks and gray relevance analysis for predicting the RUL of an induction motor. Additionally, bayesian neural network (BNN) (Zhang, Zhang, Wang, Dui, & Chen, 2024), bidirectional gated recurrent unit (BiGRU) (She & Jia, 2021), and other deep learning model have been applied for equipment RUL prediction (Deng, Du, Jia, Zhao, & Xie, 2020; Jia, Deng, Lv, Du, & Xie, 2022).

However, most deep learning-based methods assume identical data distribution across the source domain (used for training) and the target domain (used for testing). Equipment typically operates under varying conditions, and environmental factors such as temperature, humidity, load, and other parameters cause differences in the distribution of equipment operating state data, resulting in domain shifts. The prediction performance of models trained under a single operating condition decreases when applied to new operating conditions (Dong, Zhang, Liu, Wang, & Wang, 2024; Wang et al., 2023). Furthermore, it is costly and time-consuming to obtain labeling information corresponding to operational state data in industrial equipment (Liang, Wang, Jiang, Li, & Zhang, 2023). An unsupervised domain adaptation model, an essential branch of transfer learning, does not require labeling information about the target domain and addresses this challenge through acquired shared features from the domain, thereby minimizing the distribution differences between domains.

Unsupervised domain adaptation methods usually use adversarial training or metric-based approaches to adjust the distribution of features across the domains (Cheng, Kong, Chen, Wang, & Wang, 2021; da Costa, Akçay, Zhang, & Kaymak, 2020; Deng, Lv, Huang, & Du, 2023). da Costa et al. (2020) used a model combined with the domain adversarial to reduce the differences in feature distributions between domains and validate its performance using a publicly available dataset. Ye and Yu (2022) introduced a hybrid adversarial domain adaptation network addressing equipment RUL prediction problem. This network combines a domain discriminator with an unsupervised feature alignment metric to achieve domain adaptation. Zhu, Chen, and Shen (2020) designed a transferable network that incorporates the maximum mean discrepancy (MMD) technique for aligning domain feature distributions through a fully connected layer, thereby enhancing the accuracy of RUL prediction. Moreover, metric learning (Ding, Jia, Miao, & Huang, 2021), transfer component analysis (Mao, He, & Zuo, 2019), and other approaches have solved the problem of unsupervised domain adaptation for RUL prediction of equipment (Deng, Huang, Du, Li, Zhao, & Lv, 2021).

Most existing unsupervised domain adaptation models usually apply an unsupervised domain adaptation strategy to the deep features extracted from the last few layers to capture the global shared features of both domains, ignoring the domain's intrinsic properties (Hu et al., 2022). For example, in the RUL prediction of an aircraft engine, the engine temperature and rotational speed under different operating conditions are the engine's inherent properties, i.e., the domain's private properties. Forcing global feature alignment will lack strong cross-domain representation if each domain-specific feature is not considered. The model's capability to extract domain-invariant features can be enhanced by distinguishing and dividing each domain's private and shared features. Hu et al. (2022) used a transfer learning network for RUL prediction based on deep feature disentanglement, effectively dividing the domain's private and shared features. Considering domain-specific knowledge, Wang et al. (2023) developed a network for feature division and tendency retention (DR-Net) to estimate battery capacity based on a publicly available dataset. Existing models (Hu et al., 2022; Wang et al., 2023) divide the domain's private and shared features through explicit modeling methods. However, domain-specific features (i.e., domain private features) still remain in these shared features representation, which affects the performance of the model.

Additionally, in the design of feature extraction networks, many existing models often ignore the positional information between sensors deployed on equipment. This neglect may significantly affect the

model performance because of the different importance of the data collected by sensors at different locations. Moreover, in terms of shared feature extraction of domains, many networks often ignore the fact that operational status data from different health state (HS) of the equipment exhibit varying data distributions. Using only a single-domain adaptation method could obscure fine-grained features within subdomains associated with different equipment degradation levels, thereby reducing the accuracy of RUL prediction. Both Ding, Jia, and Cao (2021) and Zhang et al. (2023) considered multiple health state of the equipment and constructed corresponding subdomain adaptation models for equipment RUL prediction. These models (Ding, Jia, & Cao, 2021; Zhang et al., 2023) divide each subdomain in advance based on RUL labels without considering the specific features of each degradation stage of the equipment, which compromises model robustness. The relationship between subdomain division and RUL prediction is often ignored in unsupervised subdomain adaptation for RUL prediction.

To address this challenge, we design a novel unsupervised subdomain adaptation model to predict the RUL of equipment. Specifically, the proposed model contains four components: a feature extractor, a feature similarity component, a domain adaptation component, and a multi-task module. The feature extractor is developed to obtain private and shared features from domain data. It primarily includes a feature extraction component that utilizes a dynamic fusion network for multi-scale information and a private feature extraction component built from a dynamically fused bi-directional GRU network. The feature similarity component employs a cross-attention mechanism to reconstruct features based on the extracted shared and private features. The domain adaptation component incorporates a global domain adaptation module and a subdomain adaptation module based on MMD and local maximum mean discrepancy (LMMD), allowing the model to generalize learned information from the source to the target domain. Additionally, the multi-task module integrates a health assessment module and an RUL predictor, enabling simultaneous subdomain division and RUL prediction. The health assessment module evaluates the equipment's health state and divides the subdomains according to the health assessment results. The RUL predictor, which contains several dense layers, predicts the equipment's RUL.

This paper's main contributions are outlined below:

- (1) Considering sensor spatial information, this paper dynamically fuses the data's temporal information and the sensor's spatial information based on the temporal convolutional network and graph convolutional network to improve the model performance.
- (2) In the feature similarity module, we apply the cross-attention mechanism to effectively constrain the constructed private and shared features, strengthening the model's capability to disentangle the domain's private and shared features.
- (3) Based on the multi-task learning strategy, this paper simultaneously optimizes both subdomain division and RUL prediction to enhance the model's robustness.

The subsequent sections are presented below. The problem statement and our proposed model are described in Section 2. A description of the used dataset is provided in Section 3. The detailed parameter configuration for our model is presented in Section 4. Section 5 displays the experimental analysis. In Section 6, we discuss the conclusions and future work.

2. Problem formulation and our model

2.1. Research problem formulation

To clearly state the unsupervised subdomain adaptation problem studied in this paper, we make the following assumptions about the corresponding variables: $D_s = \{x_i^s, y_{ti}^s, y_{ci}^s\}_{i=1}^{n_s}$ and $D_t = \{x_i^t\}_{i=1}^{n_t}$ are denoted as labeled source domain and unlabeled target domain, respectively, where x_i^s and x_i^t describe the source and target domain, respectively, y_{ti}^s and y_{ci}^s represent respectively the corresponding source domain RUL

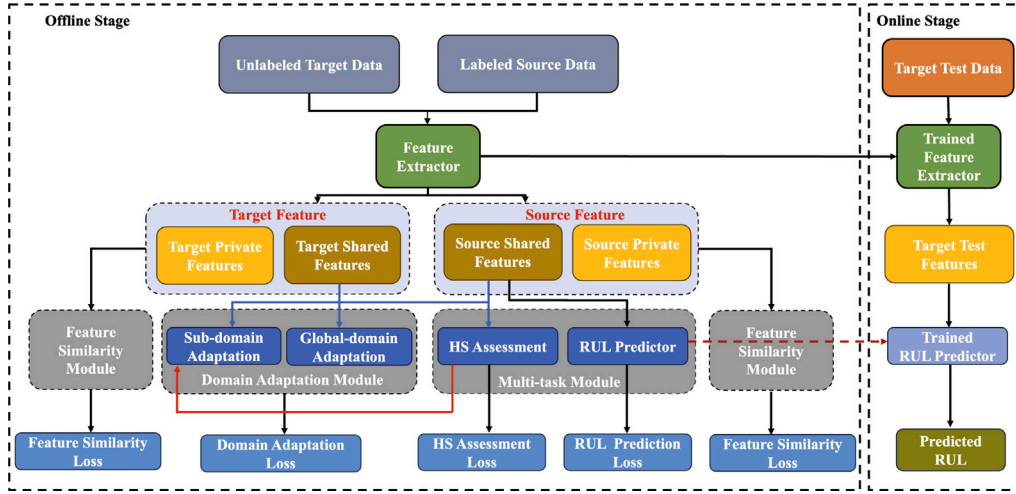


Fig. 1. Design of the proposed unsupervised subdomain adaptation model.

label and HS assessment label, the amount of data corresponding to the source and target domain is denoted by n_s and n_t , respectively. $p_s(\{x_i^s\}_{i=1}^{n_s})$ and $p_t(\{x_i^t\}_{i=1}^{n_t})$ describe the marginal distribution for source and target domain, respectively. There is a discrepancy in the marginal distributions of the two domains, i.e., $p_s(\{x_i^s\}_{i=1}^{n_s}) \neq p_t(\{x_i^t\}_{i=1}^{n_t})$. Moreover, there are differences in the distribution of conditions between domains, i.e., $p_s(y_c^s | x^s) \neq p_t(y_c^t | x^t)$, where y_c^t represents the predicted equipment's health state. $x^s \in \{x_i^s\}_{i=1}^{n_s}$, $x^t \in \{x_i^t\}_{i=1}^{n_t}$, $y_c^s \in y_c$, and $y_c^t \in y_c$. y_c describes the equipment's total number of health state. Relying solely on a global domain adaptation method to address distributional differences could obscure the fine-grained information of the subdomains that correspond to various health state of equipment, resulting in a lower RUL prediction accuracy. To address these issues, an innovative RUL prediction model is proposed. The proposed model aims to reduce the discrepancy between marginal and conditional distributions across domains, thereby improving its generalization capabilities.

2.2. The proposed model

Fig. 1 illustrates the structure of our proposed model, which is mainly divided into online and offline stages. Specifically, during the offline phase, the feature extractor captures private and shared features individually for the source and target domains. The shared features from the source and target domains are merged and subsequently transmitted to the domain adaptation component for computing the adaptation loss, where the domain adaptation component is composed of two subcomponents: sub-domain and global domain adaptation. Sub-domains of each domain are divided based on the results of the equipment HS assessment within the multi-task module. The target domain shared features utilize the HS assessment to generate subdomain pseudo-labels. The source domain shared features perform HS assessment and calculate the HS assessment loss. Furthermore, the shared features of source domain are used as inputs to the RUL predictor of the multi-task module while the RUL regression loss is computed. A new feature similarity component is designed for calculating the similarity between restructured domain features (i.e., the combination of private and shared features) and domain features (i.e., features that are not divided into private and shared features). During the online phase, the equipment's RUL is predicted utilizing the test data according to the trained feature extractor and RUL predictor components.

Different from the previous models (Ding, Jia, & Cao, 2021; Zhang et al., 2023), the proposed model does not require pre-defined subdomains; instead, it divides the subdomains according to the information of the data. Our model incorporates a feature similarity component to more effectively constrain the domain's private and shared features.

2.3. Detail of the four components

The details of the proposed model's four components are described below: the feature extractor, the feature similarity component, the domain adaptation component, and the multi-task module.

2.3.1. The feature extractor

The feature extractor contents two main subcomponents, the feature extraction component $F(\cdot)$ and the private feature extraction component $F_p(\cdot)$ (Yan, Ye, He, & He, 2024). As illustrated in Fig. 2, $F(\cdot)$ and $F_p(\cdot)$ obtain domain features and private features utilizing the domain data. Each domain's shared features f_c are acquired through removing its private features within its domain features. The shared features can be written as Yan et al. (2024):

$$f_c = F(x_i) - F_p(x_i) \quad (1)$$

where the feature extraction module extracts domain features $F(x_i)$ using a Multi-scale Feature Fusion Networks (MFFN). The $F_p(x_i)$ is utilized to extract private features based on a dynamically weighted bidirectional GRU network.

(1) The feature extraction component. As depicted in Fig. 3, the MFFN stacks four Multi-information Dynamically Weighted Fusion Networks (MDWFN) with a multiscale learning strategy. The features extracted by each MDWFN at different scales are fused by concatenation operation and transition module layers, and the extracted features are dynamically output through weight vectors. The MDWFN and transition module layers (TML) are elaborated on below.

The MDWFN mainly comprises Residual Temporal Convolutional Networks (RTCN) and Graph Convolutional Networks (GCN). The structure of the MDWFN is presented in Fig. 4. Based on the input data, the RTCN and GCN extract the temporal and spatial features, respectively. The extracted features are combined, and the results are outputted by dynamic fusion with weight vectors. The whole feature extraction process can be written as:

$$F(x_i) = w * \text{Con}(f_{rtcn}(x_i), f_{gcn}(x_i)) \quad (2)$$

where w denotes the weight vectors; $\text{Con}(\cdot)$ is the concatenation operation. $f_{rtcn}(\cdot)$ and $f_{gcn}(\cdot)$ describe the features extracted by RTCN and GCN based on the input data x_i , respectively. The details of RTCN and GCN will be given below.

RTCN is a variant of temporal convolutional networks (TCNs), which flexibly captures information in data by employing dilated causal convolution techniques and residual connectivity strategies. The network details of RTCN are described in Fig. 5, which mainly comprises three layers of TCN modules. Batch normalization is applied before

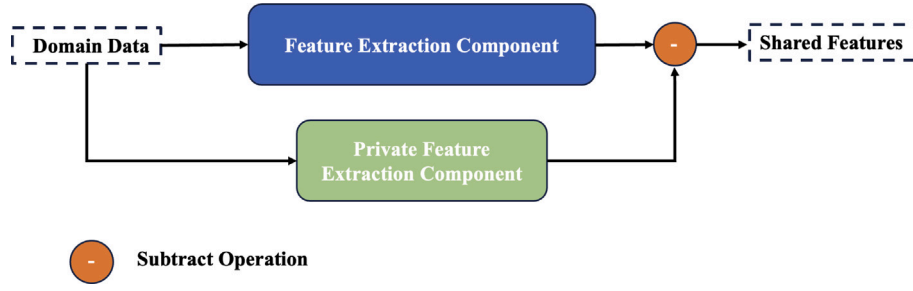


Fig. 2. The feature extractor framework (Yan et al., 2024).

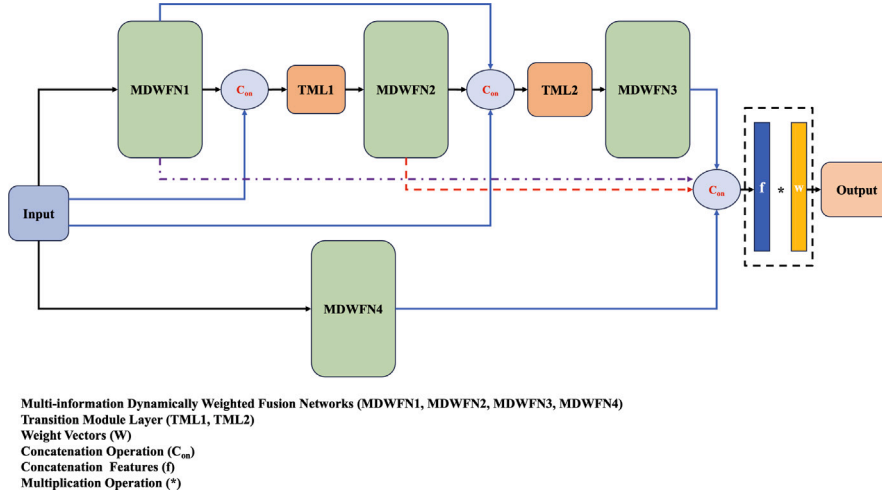


Fig. 3. The MFFN framework.

the activation function in the final TCNs module. Additionally, a one-dimensional convolutional network (Conv1D) is utilized by our model to address dimensional mismatches within the residual connections. The details about TCN can be found in Zhuang, Jia, Ding, and Ding (2021). GCN can capture non-Euclidean spatial dependencies in multi-sensor data and is widely used in PHM tasks (Wang, Cao, Xu, & Liu, 2022). Therefore, this paper employs GCN to capture spatial features from sensor data based on the adjacency matrix between the sensors and the data (Wang et al., 2022).

In the MFFN model, a Transition Module Layer (TML) is designed to regulate the dimensionality of the features, thus controlling the model's complexity (Ding, Li, Mao, Meng, & Shen, 2023). The TML connects two adjacent MDWFN modules, as shown in Fig. 3. The TML comprises a normalization, a convolutional, and a pooling layer (Yan et al., 2024).

(2) The private feature extraction component. As depicted in Fig. 6, this paper uses a Multilayer Bidirectional Gated Recurrent Unit (MBiGRU) network and GCN dynamic information fusion model to obtain private features for each domain. Specifically, the private feature extraction component combines the temporal features acquired using the MBiGRU with the spatial features extracted by the GCN. Then, it multiplies them with the weight vector to produce the final result. Among them, BiGRU can bi-directionally comprehensively capture both past and future information based on the sensor data (She & Jia, 2021). GCN effectively extracts the structural information of the data.

2.3.2. The feature similarity component

The feature similarity component proposes a cross-attention mechanism model based on feature reconstruction (see Fig. 7). As described in Fig. 7, the private features, shared features, and domain features are operated for feature transformation, respectively, i.e., f_{temp1} , f_{temp2} , f_{temp3} . Specifically, after the concatenation operation, f_{temp1} and f_{temp2}

are merged through feature combination. Subsequently, they are integrated with the weights generated by f_{temp3} after applying the sigmoid activation function, enabling cross-attention. Finally, the results of feature reconstruction ($Score$) are obtained based on the cosine similarity method. This process can be written as:

$$Score = Sim(Con(f_{temp1}, f_{temp2}) * Sig(f_{temp3}), f_{temp3}) \quad (3)$$

where $Sim(\cdot)$ indicates the cosine similarity method. $Sig(\cdot)$ describes the sigmoid activation function.

2.3.3. The domain adaptation component

Most existing domain adaptation models use a global domain alignment strategy to narrow the distribution gap between the domains, ignoring the fact that the state data collected by the equipment at different health state possess different distributions. We propose a method combining global-domain and subdomain adaptation to obtain fine-grained information.

(a) Global-domain adaptation. We employ the maximum mean difference (MMD) metrics to assess the difference in global feature distributions (i.e., marginal distributions) across the domains. The MMD is defined as:

$$MMD = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(f_{\zeta_i}^s) - \frac{1}{n_t} \sum_{i=1}^{n_t} \phi(f_{\zeta_i}^t) \right\|_H \quad (4)$$

where $\|\cdot\|_H$ describes a norm in the Regenerated Kernel Hilbert Space (RKHS), $f_{\zeta_i}^s$ and $f_{\zeta_i}^t$ (described in Eq. (1)) represent the shared features corresponding to source and target domains, respectively, $\phi(\cdot)$ means a function related to the feature map, $f_{\zeta_i}^s, f_{\zeta_i}^t \rightarrow RKHS$ (Yan et al., 2024). In this paper, the global loss function is constructed based on MMD which enables the proposed model to extract global shared features. A detailed description is given in the loss function section.

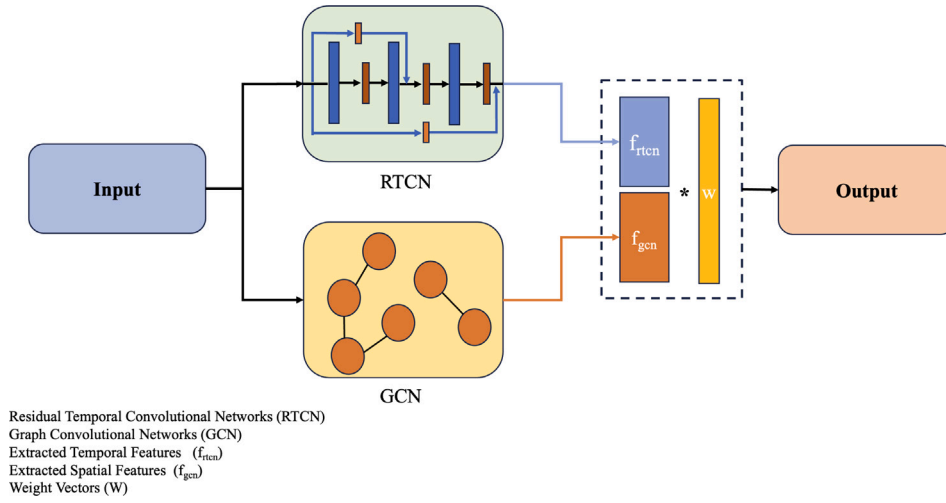


Fig. 4. Network structure of the MDWFN.

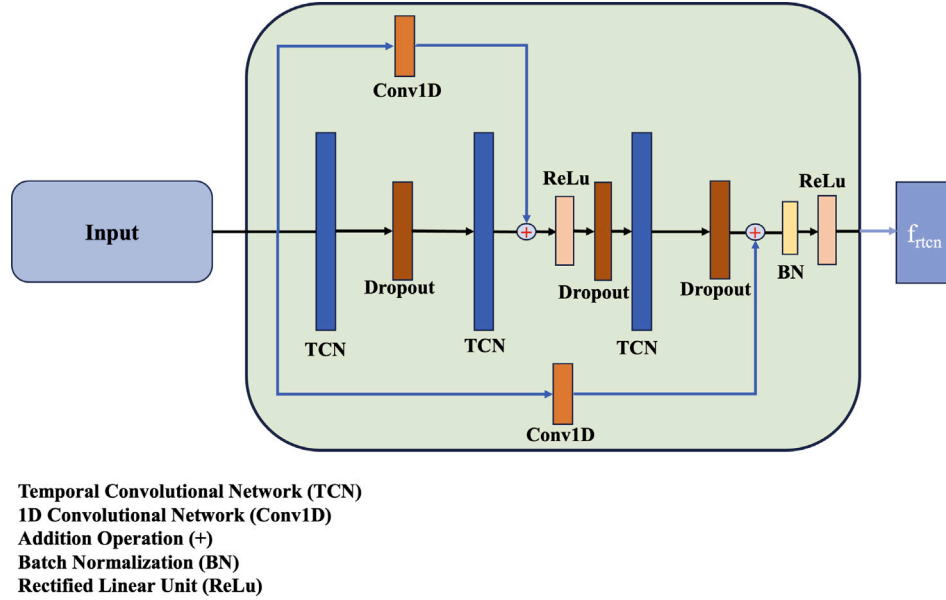


Fig. 5. Network structure of the RTCN.

(b) Sub-domain Adaptation. Most subdomain adaptation models are designed to solve classification tasks. Since the classification task has a well-defined and limited number of subdomains, the conditional probabilities of the feature distributions can be explicitly defined. However, RUL prediction describes fundamentally a regression task with an infinite variety of labeled values, resulting in zero conditional probabilities and thus making it difficult to measure differences in conditional distributions (Ding, Jia, & Cao, 2021; Zhang et al., 2023). Considering the infinite of RUL labels and the inherent similarity between different stages of the equipment degradation process, this paper proposes using the results of the HS assessment in the multi-task learning module as a sub-domain division of the degradation stages. Based on the division of subdomains, the Local Weighted Maximum Mean Difference (LWMMMD) quantifies the subdomain distribution difference across the domains. The definition of LWMMMD is as follows:

$$LWMMMD = \frac{1}{C} \sum_{c=1}^C \left\| \sum_{f_{\zeta_i}^s \in D_s} w_i^{s(c)} \phi(f_{\zeta_i}^s) - \sum_{f_{\zeta_j}^t \in D_t} w_j^{t(c)} \phi(f_{\zeta_j}^t) \right\|_H \quad (5)$$

where the C represents the overall number of subdomains, matching the quantity of equipment's health state. The weights $w_i^{s(c)}$ and $w_j^{t(c)}$

represent respectively the weights assigned to the i th shared features of source domain $f_{\zeta_i}^s$ and the j th shared features of target domain $f_{\zeta_j}^t$, belonging to class c th. $w_i^{s(c)}$ and $w_j^{t(c)}$ must satisfy $\sum_{i=1}^{n_s} w_i^{s(c)} = 1$ and $\sum_{j=1}^{n_t} w_j^{t(c)} = 1$, respectively. Additionally, for a sample $x_i^o, o \in \{s, t\}$, its weight can be defined as:

$$w_i^{(c)} = \frac{y_{ci}}{\sum_{(x_i^o, y_{ri}, y_{ci})} y_{ci}} \quad (6)$$

where y_{ci} denotes the soft-labeled vector (a one-hot encoding vector) of the equipment health state. Soft label vectors could be accurately obtained for source domain samples x_i^s with labels. For the unlabeled target domain samples x_j^t , the pseudo-labels $\hat{y}_c^t$ of the soft label vectors can be obtained by the proposed model. Furthermore, the $\hat{y}_c^t$ indicates the probability of assigning x_j^t to each c subdomain. The weight values of each sample across the domains are derived for the calculation of Eq. (5).

2.3.4. The multi-task module

The structure of our multi-task module is described in Fig. 8 and comprises two components: HS assessment component and RUL predictor component. The health state of the equipment is assessed by

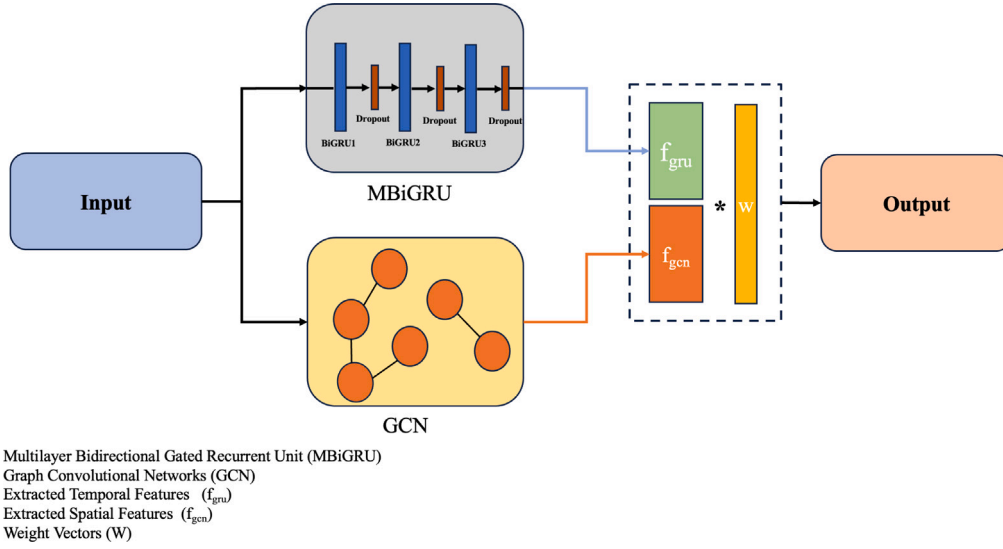


Fig. 6. Framework for the private feature extraction module.

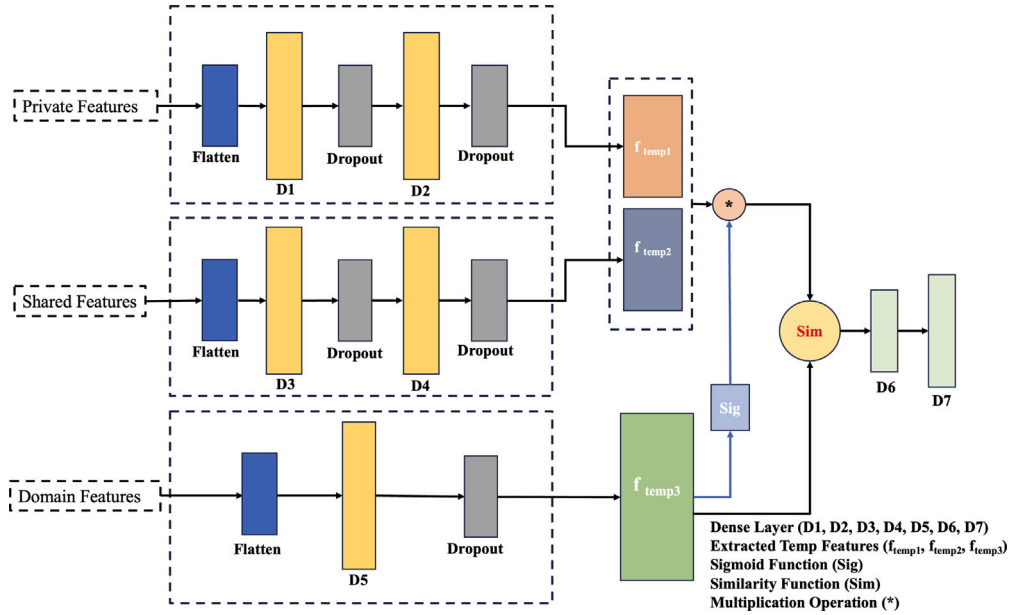


Fig. 7. The framework of the feature similarity component.

the HS assessment, while the estimated health state is used as the basis for dividing the subdomains. The HS assessment comprises a GlobalAvgPool1D (GAP) layer, three dense layers (D1, D2, D3), and two dropout layers. Additionally, the HS assessment possesses two outputs. The first part, $E_{ce}(\cdot)$, is the information obtained through the dense network, which serves for the basis for computing the adaptation loss. The second part utilizes the results $\hat{y}_c^j$ through the multi-dense network to evaluate the health state of the equipment. The RUL predictor mainly has a flattened layer, three dense layers (D4, D5, D6), and two dropout layers. Similar to the HS assessment structure, the two outputs of the RUL predictor are used as inputs for the computational domain adaptation loss and for predicting the equipment's lifetime, respectively. The mathematical description of the multi-task module is given in Eqs. (7)–(10).

$$f_{ce}^o = E_{ce}(f_c^o), o \in \{s, t\} \quad (7)$$

$$f_{re}^o = E_{re}(f_c^o), o \in \{s, t\} \quad (8)$$

$$\hat{y}_c^o = S(a(f_c^o)), o \in \{s, t\} \quad (9)$$

$$\hat{y}_r^o = \sigma(e(f_c^o)), o \in \{s, t\} \quad (10)$$

where f_{ce}^o and f_{re}^o denote the information obtained based on the first part of the HS assessment $E_{ce}(\cdot)$ and the first part of the RUL predictor $E_{re}(\cdot)$, respectively. o indicates the source s or target domain t . $\hat{y}_c^o$ represents the result of HS assessment, $a(\cdot)$ indicates the classification information obtained by D3, and $S(\cdot)$ is the softmax activation function (Bishop & Nasrabadi, 2006). Under Eq. (10), $\hat{y}_r^o$ stands for the RUL prediction value, $e(\cdot)$ means the prediction information extracted by D6, and $\sigma(\cdot)$ represents a line activation function.

2.4. The loss function of our model

The loss function L_s is illustrated in Fig. 9 and comprises the below function.

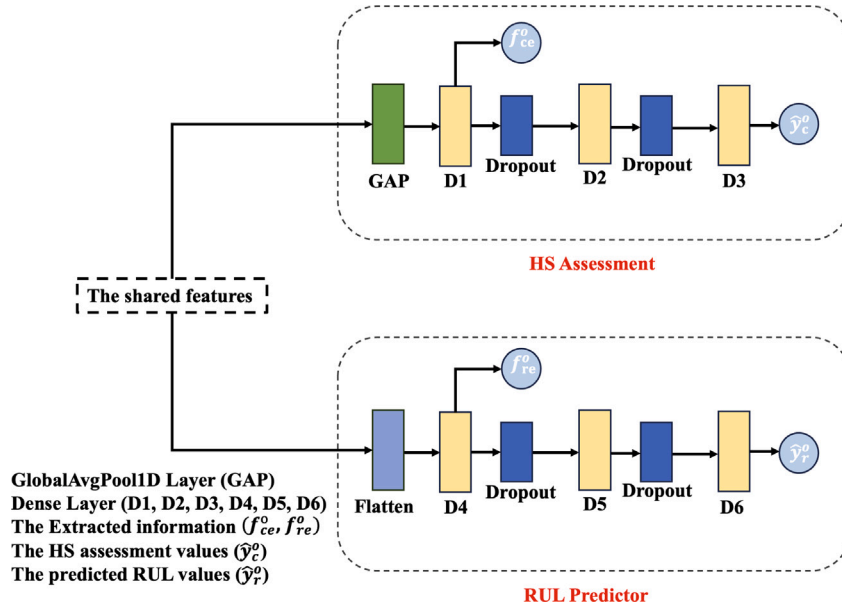


Fig. 8. The framework of the multi-task module.

(1) Minimize the feature reconstruction loss L_{re}^s, L_{re}^t for each domain based on domain features, private features and shared features;

(2) Minimize the HS assessment result error L_c and the RUL prediction error L_r using shared features from the source domain;

(3) Minimize the global domain differences L_{M1} and the subdomain domain differences L_{LM1}, L_{LM2} in the information extracted from both domains;

(4) Maximize the disparity L_{M2}^s, L_{M2}^t across private and shared features within each domain, respectively;

$L_{re}^s, L_{re}^t, L_c, L_r$: We construct a feature reconstruction loss based on the score from the feature similarity component. In the multi-task module, we compute the loss for our model's multi-task according to the source domain. The HS assessment and RUL prediction corresponds respectively to classification and regression tasks (Yan, He, & He, 2023). Therefore, we optimize the model parameters using cross-entropy classification loss for the HS assessment and mean square error (MSE) regression loss for the RUL prediction. They can be written as:

$$L_{re}^o = \frac{1}{n_f} \sum_{i=1}^{n_f} (1 - score_i), o \in \{s, t\} \quad (11)$$

$$L_c = -\frac{1}{n_b} \sum_{i=1}^{n_b} \sum_{k=1}^C y_{cik}^s \ln \hat{y}_{cik}^s \quad (12)$$

$$L_r = \frac{1}{n_b} \sum_{i=1}^{n_b} (\hat{y}_{ri}^s - y_{ri}^s)^2 \quad (13)$$

where n_f and n_b describe the feature count and batch count in the source domain, respectively. C represents the total quantity of HS classes. y_{cik}^s and $\hat{y}_{cik}^s$ refers to the true and predicted health state of the equipment, respectively. y_{ri}^s and $\hat{y}_{ri}^s$ represent the true and predicted values of RUL, respectively.

L_{M1}, L_{LM1}, L_{LM2} : The domain adaptation component, which includes subdomain and global domain adaptation, focuses on obtaining shared features from the domain. The MMD and LWMMMD (as mentioned in Section 2.3.3) are utilized to quantify the differences in the global distributions and subdomain distributions between the domains. The global adaptation module is employed to acquire global domain-invariant features by leveraging the global information from both domains. Additionally, we utilize the subdomain adaptation module to extract fine-grained shared features. In the proposed model training

process, the formulation of L_{M1}, L_{LM1}, L_{LM2} can be written as:

$$L_{M1} = \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta ei}^s, f_{\zeta ej}^s) + \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta ei}^t, f_{\zeta ej}^t) - \frac{2}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta ei}^s, f_{\zeta ej}^t) \quad (14)$$

$$L_{LM1} = \frac{1}{C} \sum_{c=1}^C \left[\sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{s(c)} w_j^{s(c)} K(f_{\zeta ei}^s, f_{\zeta ej}^s) + \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{t(c)} w_j^{t(c)} K(f_{\zeta ei}^t, f_{\zeta ej}^t) - 2 \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{s(c)} w_j^{t(c)} K(f_{\zeta ei}^s, f_{\zeta ej}^t) \right] \quad (15)$$

$$L_{LM2} = \frac{1}{C} \sum_{c=1}^C \left[\sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{s(c)} w_j^{s(c)} K(f_{\zeta ei}^s, f_{\zeta ej}^s) + \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{t(c)} w_j^{t(c)} K(f_{\zeta ei}^t, f_{\zeta ej}^t) - 2 \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} w_i^{s(c)} w_j^{t(c)} K(f_{\zeta ei}^s, f_{\zeta ej}^t) \right] \quad (16)$$

where $f_{\zeta e}^o = \{f_{\zeta ei}^o\}_{i=1}^{n_b}, o \in \{s, t\}; f_{re}^o = \{f_{rei}^o\}_{i=1}^{n_b}, o \in \{s, t\}$ and $f_{ce}^o = \{f_{cei}^o\}_{i=1}^{n_b}, o \in \{s, t\}$ refer to the information obtained by the proposed model from different network layers based on source and target domains, respectively. n_b indicates the quantity of samples according to source and target domain. $K(\cdot, \cdot)$ describes a gaussian radial basis kernel function (Hu et al., 2022).

L_{M2}^s, L_{M2}^t : The private feature extractor aims to disentangle the domain's private and shared features through increasing the distribution differences, which means increasing the L_{M2}^s and L_{M2}^t losses during the proposed model training. Inspired by Hu et al. (2022), L_{M2}^s and L_{M2}^t also employ the MMD method. The equation is denoted as:

$$L_{M2}^s = \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta i}^s, f_{\zeta j}^s) + \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{pi}^s, f_{pj}^s) - \frac{2}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta i}^s, f_{pj}^s) \quad (17)$$

$$L_{M2}^t = \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta i}^t, f_{\zeta j}^t) + \frac{1}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{pi}^t, f_{pj}^t) - \frac{2}{n_b^2} \sum_{i=1}^{n_b} \sum_{j=1}^{n_b} K(f_{\zeta i}^t, f_{pj}^t)$$

Algorithm 1 Training procedure of the proposed model

Input: $D_s = \{x^s, y_r^s, y_c^s\}$; $D_t = \{x^t\}$; Hyperparameter $\theta, \alpha, \beta, \gamma, \delta, \mu$;
 The learning rate η ; The maximum epoch $E_{\max}$; Batch size n_b ;
 Initialization model parameters θ ;

Output: the optimal parameters θ

```

1:  $i_{ep} \leftarrow 0$ ; / * The index of the epoch. * /
2: while  $i_{ep} < E_{\max}$  do
3:   for  $n_b$  do
4:     Feed  $x^s$  into the proposed model, obtain shared features  $f_\zeta^s$ , private features  $f_p^s$ , extracted information  $f_{\zeta e}^s, f_{re}^s, f_{ce}^s$ , predicted RUL  $\hat{y}_r^s$  and HS  $\hat{y}_c^s$ ;
     Feed  $x^t$  into the proposed model, obtain shared features  $f_\zeta^t$ , private features  $f_p^t$ , and extracted information  $f_{\zeta e}^t, f_{re}^t, f_{ce}^t$ ;
5:     Calculate  $L_r(y_r^s, \hat{y}_r^s), L_c(y_c^s, \hat{y}_c^s), L_{re}(f_{re}^s, \hat{f}_{re}^s), L_{re}(f_{re}^t, \hat{f}_{re}^t),$ 
 $L_{M1}(f_{\zeta e}^s, f_{\zeta e}^t), L_{LM1}(f_{ce}^s, f_{ce}^t), L_{LM2}(f_{re}^s, f_{re}^t), L_{M2}^s(f_{\zeta}^s, f_p^s),$ 
 $L_{M2}^t(f_{\zeta}^t, f_p^t)$ ;
6:     Update the model parameters  $\theta$  with the back-propagation operation and Adam optimizer to search the minimize the whole loss function  $L_s(\theta)$ ;
7:   end for
8:    $i_{ep} \leftarrow i_{ep} + 1$ ;
9: end while
10: return the  $\theta$  and stop.

```

(18)

where $f_\zeta^o = \{f_{\zeta i}^o\}_{i=1}^{n_b}, o \in \{s, t\}$ and $f_p^o = \{f_{pi}^o\}_{i=1}^{n_b}, o \in \{s, t\}$ denote the shared and private features extracted through the proposed model based on domain data.

By combining the optimization objectives mentioned above, the loss function of the proposed model is denoted as:

$$L_s = \theta L_r + \alpha L_c + \beta(L_{re}^s + L_{re}^t) + \gamma(L_{M1} + \delta L_{LM1} + \delta L_{LM2}) - \mu(L_{M4}^s + L_{M4}^t) \quad (19)$$

where $\theta, \alpha, \beta, \gamma, \delta$ and μ represent hyper-parameters. The proposed model's parameters are optimized by minimizing Eq. (19). The updated formula for the model parameter θ can be expressed as:

$$\theta \leftarrow \theta - \eta \frac{\partial L_s}{\partial \theta} \quad (20)$$

where the proposed model's parameters are optimized and converged through an Adam optimizer and the back-propagation operation (Ghorvei, Kavianpour, Beheshti, & Ramezani, 2023). η denotes the learning rate. Algorithm 1 illustrates the steps to train the optimal parameters based on L_s . The trained feature extractor is integrated with the trained RUL predictor to perform the RUL prediction according to test data.

3. Dataset description

The used datasets, the data preprocessing methods, and the label construction are described in this section.

3.1. Data introduction and preprocessing

The Commercial Modular Aeronautical Propulsion System Simulation (C-MAPSS) dataset is a multi-sensor data collected by simulating the engine degradation process, which has been widely used in the PHM research field. In this paper, we select 14 sensor data from the C-MAPSS dataset to validate our model's performance (Saxena, Goebel, Simon, & Eklund, 2008). Additionally, the positional information of the sensors (i.e., the adjacency matrix in the GCN network) is obtained through the cosine similarity between the collected data of each sensor (Wang et al., 2022). The details dataset and data preprocessing method can be obtained from Wang et al. (2022), Yan et al. (2023), Zhu et al. (2020).

3.2. RUL label and subdomain label (HS label)

Based on previous studies (Yan, He, & He, 2022; Yan et al., 2023), this paper uses the piecewise linear function to determine each equipment's RUL labels in the source domain and the change point identification method to determine the HS labels (i.e., subdomain labels). To effectively accomplish the model training and the domain knowledge transfer, we further scale the RUL label to between 0 and 1 (Yan et al., 2024). Additionally, according to the data characterization and change point identification methods, we classify the health state of each equipment in the source domain into five states. More details can be found in Yan et al. (2023).

4. Parameter configuration

We determine the optimal parameters of the proposed model through pre-experiments. Table 1 presents the detailed structure and parameter configuration. In our model, the feature extraction component includes multiple MDWFN layers and TML layers, each layer with different parameters, as detailed in the table. The private feature extraction component comprises multiple BiGRUs and GCNs with different units. The feature similarity component consists of multiple dense layers and utilizes cross-attention and cosine similarity methods to obtain the reconstructed features similarity scores. Additionally, the multi-task module consist of multiple dense layers with different numbers of neurons, activation functions, and dropout layers, respectively.

In the training phase, the learning rate of the Adam optimizer equals 0.0001 (Kingma & Ba, 2024). Our model uses a batch size of 128 (Yan et al., 2024). A maximum of 200 epochs is used for training our model. For the C-MAPSS datasets, the sliding window size is configured to 50. Additionally, the hyper-parameters in our loss function are configured as $\alpha = 0.001, \theta = 10, \beta = 0.01, \gamma = 0.1, \mu = 0.0005, \delta = 0.5$ using a cross-validation approach (Wong & Yeh, 2019).

5. Experiment result

The evaluation metrics and the experimental results from comparisons with relevant existing models are presented.

5.1. Performance evaluation metrics

Three metrics commonly used in the field of PHM are employed to assess the effectiveness of our model, i.e., mean absolute error (MAE), root mean square error (RMSE), and penalty score function (PSF) (Saxena et al., 2008). Different from the metrics RMSE and MAE, the PSF considers that delayed predictions usually cause more significant losses than early predictions. Moreover, the PSF applies different penalties to prediction results. Their formulas are denoted as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{ri} - \hat{y}_{ri})^2} \quad (21)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{ri} - \hat{y}_{ri}| \quad (22)$$

$$PSF = \frac{1}{n} \sum_{i=1}^n PSF_i, \quad PSF_i = \begin{cases} e^{-\frac{y_{ri} - \hat{y}_{ri}}{13}} - 1, & y_{ri} - \hat{y}_{ri} < 0 \\ e^{\frac{y_{ri} - \hat{y}_{ri}}{10}} - 1, & y_{ri} - \hat{y}_{ri} \geq 0 \end{cases} \quad (23)$$

where y_{ri} and $\hat{y}_{ri}$ represent the actual and predicted RUL values, respectively. n denotes the number of equipment operational cycles. An optimal RUL prediction model has lower RMSE, MAE, and PSF values.

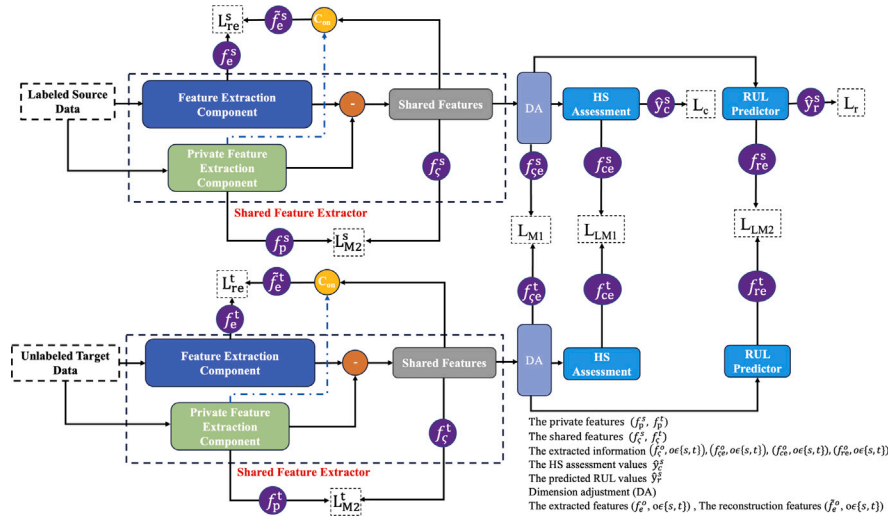


Fig. 9. The loss function structure.

Table 1
Parameter configure.

Component name	Configuration name	Layer name	Value
Feature Extraction Component	MDWFN1	TCN	filters = 8, kernel_size = 3, dilation_rate = 1
		Dropout	rate = 0.2
		Conv1D	filters = 16, kernel_size = 1
		GCN	filters = 8
	MDWFN2	TCN	filters = 16, kernel_size = 3, dilation_rate = 2
		Dropout	rate = 0.2
		Conv1D	filters = 16, kernel_size = 1
		GCN	filters = 16
Private Feature Extraction Component	TML1, TML2	Conv1D	filters = 16, kernel_size = 1
		AVgPool1D	pool_size = 2, strides = 1, padding = same
		TCN	filters = 32, kernel_size = 3, dilation_rate = 3
		Dropout	rate = 0.2
	MDWFN3	Conv1D	filters = 16, kernel_size = 1
		GCN	filters = 32
Feature Similarity Component	MDWFN4	TCN	filters = 64, kernel_size = 3, dilation_rate = 4
		Dropout	rate = 0.2
		Conv1D	filters = 16, kernel_size = 1
		GCN	filters = 64
	HS Assessment	BiGRU1	units = 8, activation = Tanh
		BiGRU2	units = 16, activation = Tanh
		BiGRU3	units = 32, activation = Tanh
		Dropout	rate = 0.2
Multi-task Module	RUL Predictor	D1	units = 64, activation = Relu
		D2	units = 32, activation = Relu
		D3	units = 5, activation = Softmax
		Dropout	rate = 0.2
	RUL Predictor	D4	units = 128, activation = Relu
		D5	units = 32, activation = Relu
		D6	units = 1, activation = Linear
		Dropout	rate = 0.2

5.2. Results and comparison

5.2.1. The effectiveness of the proposed method

This paper proposes an unsupervised subdomain adaptation model that divides each domain into private and shared features. It aligns fine-grained features across shared subdomains associated with various health state, thus minimizing the negative impact of domain data distribution differences on model prediction performance. This paper visualizes and analyzes the extracted relevant features. With the task

of FD001 (source domain data) $\rightarrow$ FD003 (target domain data) as an example, we visualize these features based on t-distributed stochastic neighborhood embedding (t-SNE) and kernel density estimation (KDE) (Li, Shang, Gao, Qian, & Feng, 2022). The visualization of feature distribution results is presented in Fig. 10. As depicted in subfigure (a), there is a notable difference in the distribution between x^s and x^t collected from different operating conditions. The private features f_p^s and f_p^t for the two domains are obtained using the private feature extractor. As depicted in subfigure (b), these features exhibit a

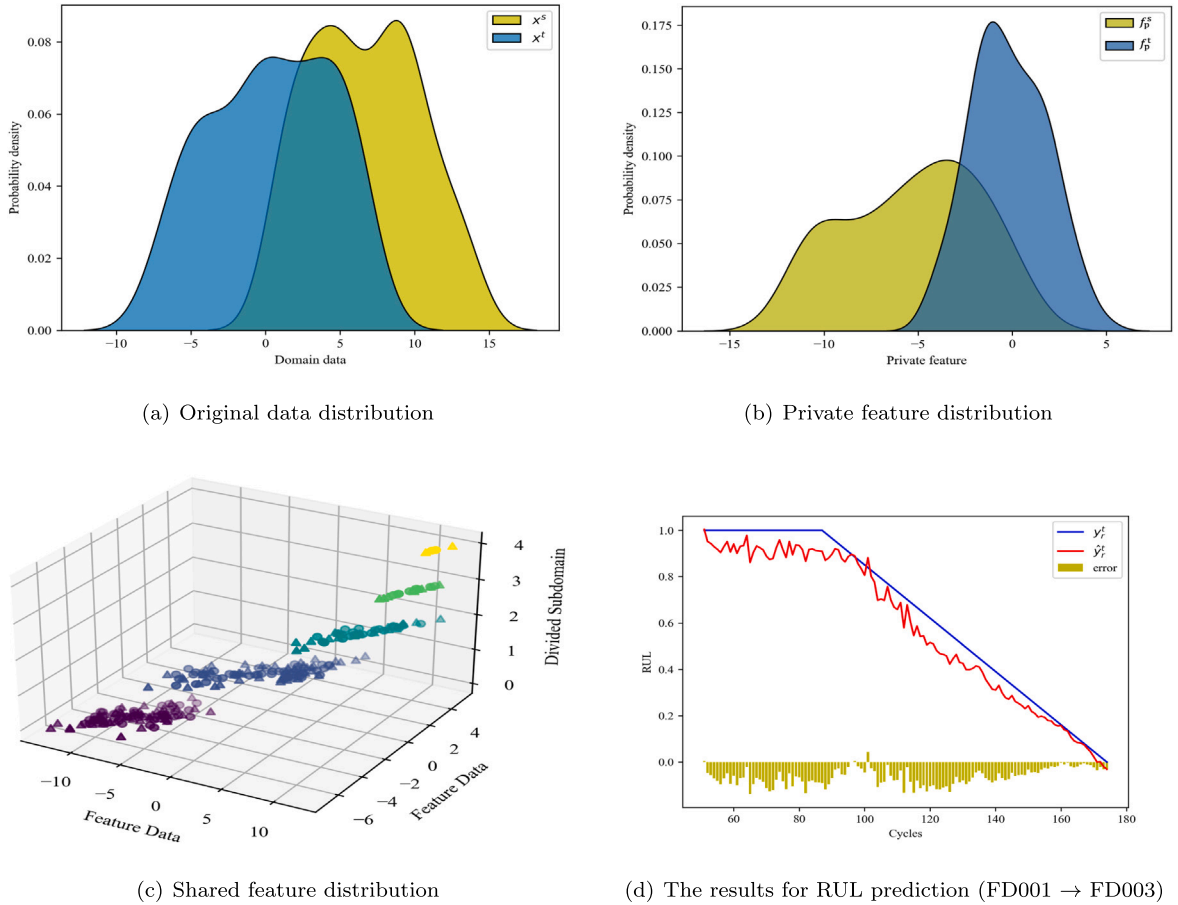


Fig. 10. Feature distribution and prediction result.

significant distribution difference. As can be seen from subfigure (c), the shared feature distributions between the domains across various subdomains exhibits significant similarity. Therefore, our model can effectively divide between each domain's private and shared features.

Additionally, as observed from subfigure (d), our model shows good performance in the transfer task of RUL prediction. The predicted RUL values $\hat{y}_r^t$ more accurately represent the true RUL values y_r^t of the equipment. Moreover, as the equipment undergoes more operating cycles, the model acquires more information regarding the equipment's operating state, yielding improved prediction accuracy in the later stages of its operational life.

5.2.2. Comparison with related works

We construct four transfer tasks utilizing publicly engine datasets (designated as transferring tasks A, B, C, and D) to verify the performance of our model. Details on constructing the transferring tasks are given in Yan et al. (2024). Moreover, this paper compares the proposed model with related works, i.e., the deep feature disentanglement with transfer learning network (DFDTLN) (Hu et al., 2022), the DRNet (Wang et al., 2023), the deep subdomain adaptive regression network (DSARN) (Ding, Jia, & Cao, 2021), and the variational auto-encoder-LSTM and local weighted deep sub-domain adaptation network (VLSTM-LWSAN) (Zhang et al., 2023). DRNet and DFDTLN consider the private features of the domains. DSARN and VLSTM-LWSAN consider the fine-grained features of the subdomains. The DFDTLN, DR-Net, DSARN, and VLSTM-LWSAN models use the parameter settings specified in their respective papers. Additionally, we establish a baseline model, referred to as Model 1, by removing the domain adaptation component of our proposed model. By removing the proposed model's

private feature extraction module and subdomain adaptation component, it is denoted as Model 2. The parameter settings of Model 1 and Model 2 are identical to those used in our model. Furthermore, the experimental results in DFDTLN and DSARN are directly referenced from experimental result in the previous paper (Yan et al., 2024).

The experimental results of the different models in the four RUL prediction of transferring tasks are presented in Tables 2–4. The tables (*RMSE*, *MAE*, and *PSF*) list the average results obtained by each model after ten runs in each test transferring task. Additionally, the best results are presented in bold, and the total average performance of each model across all of the transferring tasks is also given in the last column of each table. From Tables 2–4, it is evident that our unsupervised subdomain adaptation model outperforms other models in most tasks. On the C transfer task, the optimal model (Model 2) and the proposed model perform similar results in terms of *RMSE*, *MAE*, and *PSF* metrics. However, Model 2 exhibits poorer RUL prediction performance on the other tasks.

In terms of *RMSE* (illustrated in Table 2), our unsupervised subdomain adaptation model shows lower *RMSE* values in the transfer test tasks (except the C transfer task) compared to other comparison models. It suggests that our model outperforms the others in cross-domain RUL prediction. Compared to Model 1, our model leverages an unsupervised domain adaptation strategy to learn shared features, thereby enhancing prediction performance in the transfer task. Compared with the experimental results of Model 2 (without a private feature extraction module and subdomain adaptation strategy), the proposed model effectively removes the domain's private features and performs the alignment of the shared features at different health state through the private feature extraction module and subdomain adaptation strategy, thus improving the cross-domain performance of the model. However, accurately removing the domain's private features and identifying the boundaries

Table 2
Experimental results ($\overline{RMSE}$) for different model.

Model Name	Transfer Task				Average Result
	Task A	Task B	Task C	Task D	
Model 1	0.1569	0.1668	0.1121	0.1618	0.1494
Model 2	0.1439	0.1086	0.1042	0.1250	0.1204
DFDTLN	0.4863	0.1462	0.3550	0.1727	0.2901
DR-Net	0.2623	0.2223	0.1404	0.2223	0.2118
DSARN	0.2717	0.1077	0.1374	0.2099	0.1817
VLSTM-LWSAN	0.3262	0.1419	0.1528	0.2250	0.2115
Our Model	0.1057	0.0852	0.1081	0.0916	0.0977

Table 3
Experimental results ($\overline{MAE}$) for different model.

Model Name	Transfer Task				Average Result
	Task A	Task B	Task C	Task D	
Model 1	0.1375	0.1496	0.0864	0.1473	0.1302
Model 2	0.1257	0.0927	0.0843	0.1078	0.1026
DFDTLN	0.3861	0.1230	0.3027	0.1500	0.2405
DR-Net	0.2324	0.1882	0.1149	0.1984	0.1835
DSARN	0.2389	0.0886	0.1016	0.1849	0.1535
VLSTM-LWSAN	0.2656	0.1184	0.1140	0.1835	0.1704
Our Model	0.0882	0.0679	0.0852	0.0767	0.0795

Table 4
Experimental results ($\overline{PSF}$) for different model.

Model Name	Transfer Task				Average Result
	Task A	Task B	Task C	Task D	
Model 1	0.0137	0.0142	0.0081	0.0142	0.0126
Model 2	0.0123	0.0074	0.0072	0.0092	0.0090
DFDTLN	0.0368	0.0107	0.0301	0.0126	0.0226
DR-Net	0.0194	0.0149	0.0099	0.0156	0.0150
DSARN	0.0189	0.0073	0.0093	0.0150	0.0126
VLSTM-LWSAN	0.0241	0.0099	0.0101	0.0158	0.0150
Our Model	0.0077	0.0056	0.0076	0.0065	0.0069

across neighboring degradation stages may lead to overfitting of the model, thus reducing the generalization performance (Yan et al., 2024).

Compared with DR-Net and DFDTLN models, the proposed model considers the domain's private features and solves the subdomain adaptation problem. Experimental results indicate that our model achieves superior performance across various transfer tasks. Furthermore, compared with DSARN and VLSTM-LWSAN models, our model incorporates sensor location information into the feature extraction process, enhancing its ability to extract deep subdomain invariant features. The $\overline{MAE}$ metrics (shown in Table 3) draws the same conclusion. Additionally, the $\overline{PSF}$ (illustrated in Table 4) reflects the results achieved by utilizing different penalties to prediction. The practicability of the proposed model is further demonstrated through the PSF metrics. These tables indicate that the proposed model outperforms the majority of tasks.

5.2.3. Testing stability

To test the stability, this paper presents the performance of comparison models using box plots (Fig. 11) for the transfer task conducted on multiple equipment. Fig. 11 demonstrates that our model achieves superior RUL prediction accuracy and robustness across various transfer tasks. In contrast, the feature rebuilding technique in DFDTLN and the adversarial training in DR-Net may cause the model to exhibit unstable prediction results in different transfer tasks. Compared to the proposed unsupervised subdomain adaptation model, DSARN only considers the temporal information of the data, and the VLSTM-LWSAN model does not incorporate the sensor location information into the feature extraction, which leads to its exhibiting instability in different transfer tasks.

Table 5
The inference times among various model.

Model Name	Transfer Task B		Transfer Task D	
	Inference time	MAE	Inference time	MAE
Model1	0.1900	0.1496	0.1708	0.1473
VLSTM-LWSAN	0.1651	0.1184	0.1959	0.1835
Our Proposed Model	0.4257	0.0679	0.4253	0.0891

5.2.4. Effect of sensor location information on prediction stability

To further validate the effect of fusing sensor location information on the performance of our model, this paper compares its prediction results with those of a model that does not integrate sensor location information (the proposed model removes the GCN network module, i.e., Model 3). The comparison results, presented in Fig. 12, illustrate the distribution of prediction outcomes for the $\overline{MAE}$ metrics of RUL across various transfer tasks. The box plot (Fig. 12(a)) shows the $\overline{MAE}$ metric prediction results in distribution for multiple equipment under each task. The radar plot (Fig. 12(b)) depicts the average $\overline{MAE}$ metric values for each transfer task, visualizing the prediction performance between the two models across tasks. As can be seen from the radar plot, the proposed model shows more significant advantages in the D task with different operating conditions and different failure modes. According to Fig. 12, the proposed model shows greater stability in prediction results across different transfer tasks. This indicates that the module integrating sensor location information within the proposed unsupervised subdomain adaptation improves both prediction performance and stability.

5.2.5. Inference time comparison

We compare the inference times across various models. To guarantee fairness, all models are run ten times based on each transfer task (B and D), and the average inference time (in seconds) for these ten runs is recorded. Table 5 lists the average inference time for all models across different tasks and the corresponding mean performance metrics ($\overline{MAE}$). The table indicate that the proposed models have an average inference time of less than 0.5 s across different tasks. Compared to Model 1 and VLSTM-LWSAN, domain adaptation strategy and sensor location information are added to the proposed models, and therefore, Model 1 and VLSTM-LWSAN take less inference time on different transfer tasks. However, the proposed model performs better than Model 1 and VLSTM-LWSAN across various transfer tasks.

6. Conclusions

We introduce a novel model for predicting equipment RUL. The model comprises four main components: a feature extractor, a feature similarity component, a domain adaptation component, and a multi-task module. Firstly, the feature extractor dynamically fuses the temporal information of the data and the sensor location information to extract deeper domain private and shared features. Subsequently, these extracted features are fed into the feature similarity component and the domain adaptation component, respectively, to efficiently constrain the private features and extract the deeper subdomain shared features. The multi-task module co-optimizes subdomain division and RUL prediction. The proposed model effectively disentangles domain private and shared features and performs fine-grained feature alignment to facilitate the model's generalization capability. The model's effectiveness is validated by multiple transfer tasks on a public dataset and compared with other models.

This study involves the target domain in training the model. However, in practical applications, it is challenging to obtain information about the target domain in advance. Therefore, our future research direction is to design a new method for generalizing the model trained only based on source domain data. Additionally, we will consider the explainability of the models in subsequent works.

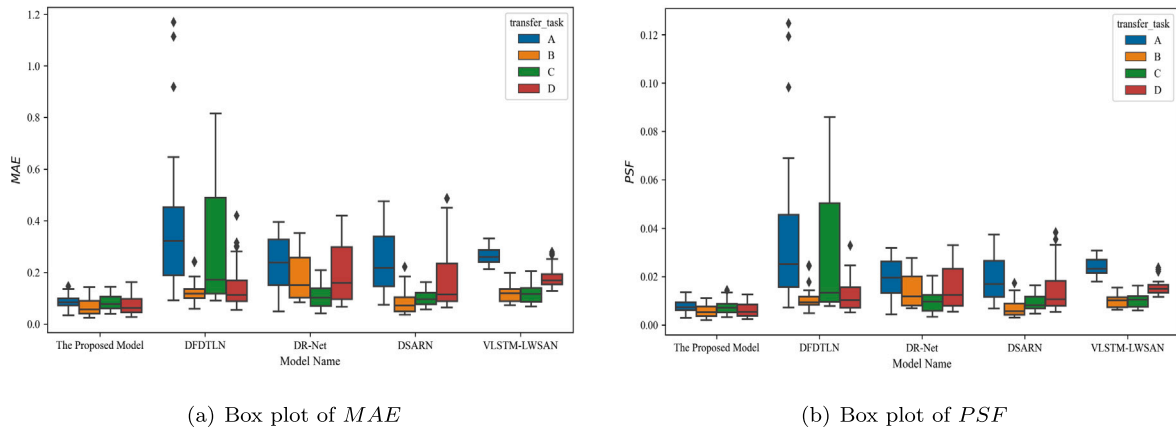


Fig. 11. The comparison of MAE and PSF across different models utilizing transfer tasks.

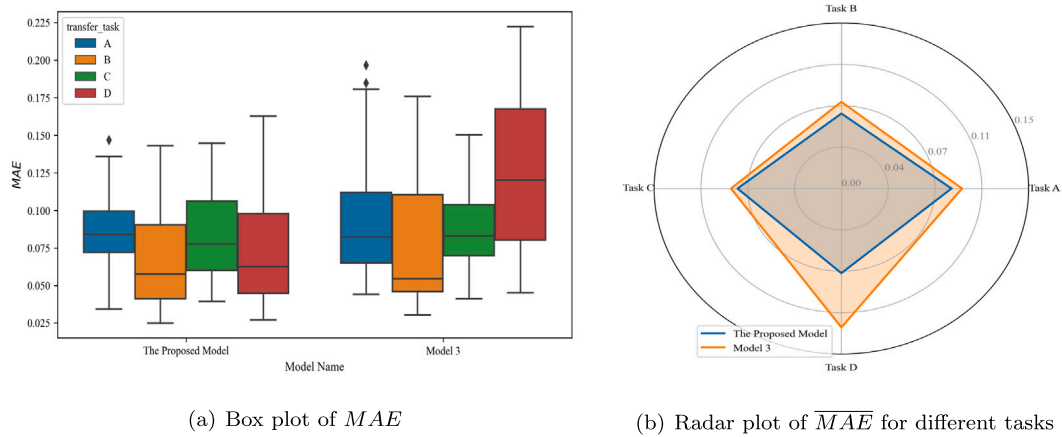


Fig. 12. The comparison of MAE across different models utilizing transfer tasks.

CRedit authorship contribution statement

Jianhai Yan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhi-Sheng Ye:** Writing – review & editing, Supervision. **Shuguang He:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Zhen He:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study are publicly available datasets.

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